

Rapid Fabrication of Haptic and Electric Input Interfaces using 4D Printed Origami

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Figure 1: Five types of shape modules as self-folded haptic and electric interfaces. (a) a joystick module and a grip module for a gamepad; (b) a press module for an interactive desk lamp; (c) a rotation module for an electronic door lock; (d) a grip module; (e) a slide module.

ABSTRACT

Origami is widely used in tangible input interfaces for diverse shapes and haptic feedback. However, the design and fabrication process of origami is labor-intensive, and building circuits on it is often unstable as they can easily be broken during iterative folding processes. We propose a fabrication method that helps users quickly design and fabricate haptic origami interfaces with electric sensing. By extending the previous 4D printing method [16], we developed a design software that can generate both a pattern of self-foldable shape and a pattern of circuitry. In the software, two kirigami hinges and five shape modules are available for users. Kirigami hinges protect the circuits from damage and achieve deformation sensing during the iterative folding process. Five shape modules have different haptic capabilities for different use cases. We summarized the fabrication process and designed three applications to validate our proposed method.

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CCS CONCEPTS

• **Human-centered computing** → **Interaction devices**; • **Applied computing** → *Computer-aided manufacturing*; • **Hardware** → *Printers*.

KEYWORDS

Rapid Prototyping, 4D Printing, Tangible Interaction, Origami

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1 INTRODUCTION

Compared to traditional screen-based interaction, tangible interaction allows users to interact with computers within the real world [6]. They have been used for various purposes, such as electronic arts [21], wearable devices [9, 15, 34], digital fabrication [26], and STEM education [11, 22]. Among them, origami, thanks to its low cost and diverse structures, has been widely applied in prototype design and manufacturing. Origami can construct structures

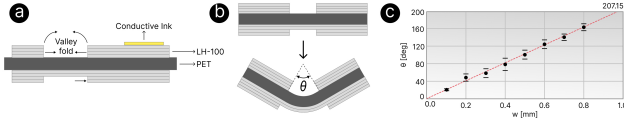


Figure 2: (a) A cross-sectional schematic image with printed ink layers; (b) process of folding of printed ink layers; (c) correlation between the folding angles and the widths of the crease; the error bars show the standard deviation, and the number next to the fitted line indicates its slope.

exceeding the strength of the material itself [7] or provide various forms of kinesthetic feedback [3].

Despite these benefits, there are still deficiencies in the design and fabrication of origami-based tangible devices. First, in design, designers often need to create a 2D pattern from a target 3D object, which is time-consuming and challenging for both novices and professionals of origami. Second, in fabrication, complex folding structures often require a large amount of manual labor. Additionally, although implementing circuits on the surface of origami could achieve user interaction with a computer via objects, the circuits are prone to damage at the creases, leading to unstable signals and a very short lifespan of electronic origami products [19, 35].

To solve these issues, we propose a fabrication method of rapidly creating tangible and interactive devices with kinesthetic feedback and electric sensing. We developed a Grasshopper-based design software that allows the user to design both a self-foldable origami shape and conductive traces on the surface, by extending the prior 4D printing method [16]. The system supports two kirigami hinges for circuitry and five shape modules for haptics. For two kirigami hinges, the tomoe pattern [14] protects the circuit from damage due to iterative folding processes, and the pogo-inspired pin we developed allows the deformation sensing of origami structures. Finally, we summarized the fabrication process and designed three applications to validate our method.

The main contributions of this work are as follows:

- A pipeline of software and hardware that can design and fabricate self-foldable tangible interfaces with haptic feedback and electric sensing;
- Five shape modules and a new circuit structure using pogo pins;
- Three applications validating the design possibility of our method.

2 RELATED WORK

2.1 4D Printing and Self-folding

Recently, 4D printing and self-folding have been applied to HCI and CG, which ease the fabrication process of 3D shapes by programmable transformation of 2D shapes with stimuli like heat and (de)hydration [23, 24, 30]. For example, fused deposition modeling (FDM) 3D printers are used to intentionally warp the 2D thermoplastic structures by heat after fabrication [1, 10, 12, 25, 27–29]. Among them, Thermorph [1] achieved self-foldable 2D origami

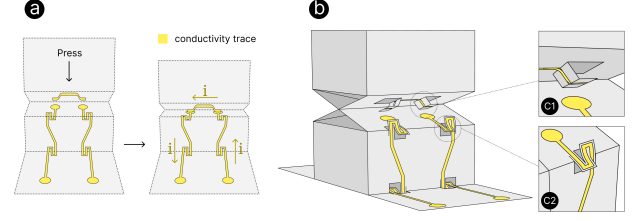


Figure 3: Kirigami hinges. (a) Signal input by circuit on and off; (b) perspective view of a button module; (c) c1 shows the pogo-inspired pin structure that we designed; c2 shows the structure of the kirigami circuit we designed to avoid damage during circuit folding.

shapes extruded from an off-the-shelf FDM printer. Also, MorphingCircuit [27] combines FDM-based 4D printing and screen printing of conductive ink to build 3D electric circuits (e.g., a lighting Christmas tree and a locomotion robot), which is regarded as the most related work to this paper. However, FDM-based 4D printing only allows for rigid shapes. Thus it is not possible to leverage the iterative mechanical deformation of origami as haptic interfaces.

In 2023, Inkjet 4D Print [16] was proposed as another approach, which self-folds complex origami shapes with a thin film (approx. 200 μm in thickness) using a UV printer. In this paper, we extended [16] in terms of two aspects: (1) we leverage the thin origami structure to achieve haptic interfaces that can iteratively fold and unfold, rather than building final shapes; (2) we combined conductive ink to achieve both shapes and electric circuits.

2.2 Flexible Electronics in HCI

HCI researchers have been applying electric circuits implemented on flexible sheets (e.g., paper, thin plastic/elastomer film, etc) to diverse interactive devices, such as flexible/stretchable displays [20, 32], customizable keyboard [2, 3], wearable [31] or cuttable sensors [18, 33], 3D objects with stacked [17] or transferred circuits [4, 8], and folded interactive objects [19]. The thin form factor of sheets is beneficial for rapid prototyping when they are combined with printing techniques. Indeed, most of the previous works adopt screen printing [8, 20, 31, 32] or inkjet printing [2–4, 8, 13, 17–19, 33]). However, when we try to steeply fold such sheets, the circuit is easy to break at the hinges. To solve this issue, Lee et al. used a kirigami structure called a tomoe pattern for origami caterpillar robots [14], which mediates the non-continuous bending at hinges with zigzag conductive traces for better stability.

In this paper, we also rely on screen printing and the tomoe pattern to achieve stable origami circuits. In addition to them, we also developed a pogo pin that detects the kinesthetic motion of origami structures. These are integrated into our design software.

2.3 Origami Haptics

While there are many efforts done in HCI and VR, here we focus on the haptic devices achieved by the origami mechanism. First, Foldio [19] folded inkjet-printed or screen-printed circuits into 3D shapes and built 3D structures with mechanical sensors and displays.

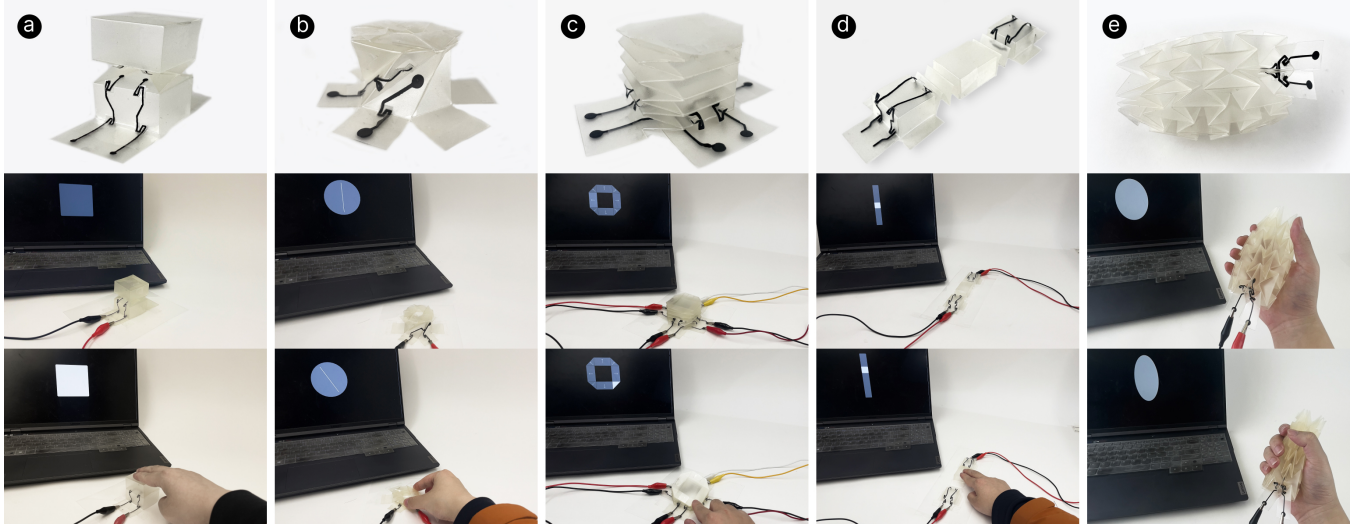


Figure 4: Shape modules. (a) Press Module; (b) Rotation Module; (c) Joystick Module; (d) Sliding Module; (e) Grip Module.

Some of the proposed structures included shape-changing and haptic interaction by folding. Similarly, Kirigami Haptic Swatches [2, 3] leveraged kirigami primitive shapes to build kinesthetic devices.

However, these prior works relied on the manual folding process, which requires more labor as the shape gets complicated. We automate the folding process of haptic interfaces by 4D printing.

3 METHOD

In this section, we detail the self-folding principles, circuit structure, signal input methods, and the shape modules enabling five different input actions.

3.1 Folding Principle

We used 4D printing to automatically fold printed planes into 3D objects, reducing manual labor costs and increasing efficiency. Previously, Inkjet 4D Print [16] introduced a method using Hakko Shrink Film 841-02 (transparent polyolefin film) as heat-shrinkable base sheets for creating 4D printed origami. As a sheet material, we used Mitsubishi Chemical PX-40S (PET, thickness: 25 μm , shrink ratio: 5% in both vertical and horizontal directions at 65 $^{\circ}\text{C}$), because the PET film was found to be more durable and therefore more suitable for iterative folding and unfolding processes than the polyolefin film used in [16], with higher rigidity and better stickiness to ink layers. We used a constant temperature water bath at 65 $^{\circ}\text{C}$ for heating.

We used the Mimaki UJF-6042 MKII as the UV printer, and Figure 2a shows the printed layer structure. We utilized one material: LH-100 (transparent) and conductive ink (Electric Paint produced by Bare Conductive [5]). LH-100 is directly printed using the UV printer, while the conductive ink is printed using screen printing after the LH-100 printing is completed. We printed four layers of LH-100 on each side, where the first two layers are for folding (in the following content, we refer to them as the M layer or V layer) and the last two for increased hardness (in the following content, we refer to them as the H layer). As a rapid manufacturing method,

the printing time for all the modules mentioned in this document is under five minutes (a total of 8 layers of LH-100). When the printed sheet is heated to 65 $^{\circ}\text{C}$, it shrinks, but the LH-100 does not, causing the paper to bend (as shown in Figure 2b).

We used the formula proposed in [16] to calculate the relationship between the folding angle and the gap width. Through calculation, we found that the folding angle θ [$^{\circ}$] equals 200 times the gap width w [mm]. To verify the accuracy of this relationship, we tested the folding angles at different gap widths. We printed 10 sets of cases, each with 8 samples of different gap widths ranging from 0.1 mm to 0.8 mm, at intervals of 0.1 mm. We measured their folding angles. As shown in Figure 2c, the material's coefficient of determination $R^2 = 0.86$, indicating a strong positive correlation. Moreover, the parameter calculated from the actual samples is 207.15, very close to the derived result of 200.

3.2 Circuit Structure

Figure 3 is a sample used to demonstrate the details of our method and explain its interaction principles. As shown in Figure 3b, this is an origami structure with a button function, featuring circuit paths made using screen printing on its surface. Multiple studies have mentioned that circuits on creases are unstable and easily broken [19, 35]. To address this issue, we route the conductive traces through tomoe patterns [14] that effectively reduces circuit deformation during folding, thus protecting the circuit (as shown in Figure 3c2). Additionally, we determine the deformation of the sample by the connectivity (connected or disconnected) of the circuits between two folded planes. To enhance the accuracy and stability of the contact detection, we designed a pogo-inspired pin structure that mimics the principle of a pogo pin. When the two surfaces come close, this improves the contact quality of the circuits on both surfaces (as shown in Figure 3c1). Figure 3a illustrates the working process of the circuit. When the button is not pressed, the circuit is in an open state. When the user presses the button, the two pins on the top connect with the circuit below, forming a

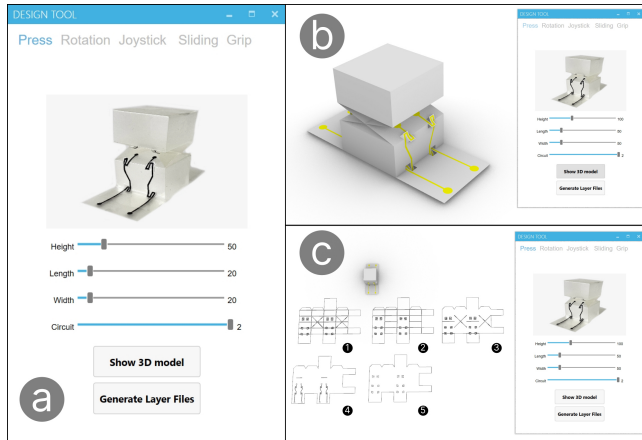


Figure 5: Design software. (a) GUI of the design plugin; (b) adjust the parameters of the module to generate the corresponding 3D preview; (c) generate files for fabrication

closed circuit. A fabricated button of this sample can be found in Figure 4a. In the following text, we uniformly refer to this type of circuit structure as a switch circuit.

3.3 Shape Module

We propose a set of basic shape modules that users can directly employ to construct more complex tangible interfaces. Detailed descriptions of each structure are as follows:

(a) Press Module: As shown in Figure 4a, this is a structure that mimics a button. This structure can be pressed and provides force feedback similar to traditional buttons, along with automatic rebound. By placing switch circuits on both sides of the module, it is possible to detect whether the press module has been pressed, thus achieving signal input. In Figure 4a, when the user presses the button and both circuits are connected, it indicates that the press module is fully pressed.

(b) Rotation Module: As shown in Figure 4b, this is a structure that mimics a knob, capable of performing angular rotation. A larger angular rotations can be achieved by stacking multiple rotation modules. Switch circuits placed on the sides of the module can detect the rotation status of the module, thus facilitating signal input. In Figure 4b, when the user rotates the button and the switch circuit changes, it indicates that the module has been rotated.

(c) Joystick Module: As shown in Figure 4c, this is a structure that mimics a joystick, allowing multi-directional signal input, supporting up to 8 directions. By placing switch circuits in each of the four directions of the module, input detection in up to 8 directions is possible. In Figure 4c, when the user presses the joystick towards the bottom right, the circuits on the bottom and right are connected, indicating that the user has moved the joystick in that direction.

(d) Sliding Module: As shown in Figure 4d, this is a structure that mimics a slider, simulating the input method of a slider, with upward or downward movement. By incorporating switch circuits in the simulated springs above and below, the slider's upward and downward movements can be detected. In Figure 4d, when the

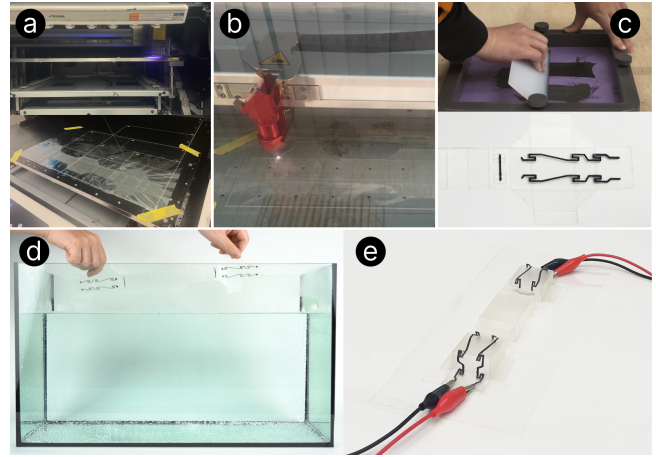


Figure 6: Fabrication process. (a) Using a UV printer to print files generated by the software; (b) using a laser cutter to remove excess material and process the kirigami structure; (c) fabricating circuits using screen printing; (d) heating in a water bath at 65 °C; (e) connecting with PCB for signal input

user slides the slider upwards and the circuit above is connected, it indicates that an upward movement of the slider has been detected.

(e) Grip Module: As shown in Figure 4e, this is a structure that mimics a grip, enabling signal input through gripping. By incorporating a switch circuit within the structure, grip signal input can be detected. In Figure 4e, when the user grips the module tightly and the switch circuit connects, it indicates that the user has gripped the module.

4 DESIGN AND FABRICATION

4.1 Design Process

Due to the repetitive nature of work in the design process, such as the fixed structure of each module which requires size adjustments for different design scenarios, we developed a Grasshopper-based plugin. Users can directly utilize our library to access five modules and adjust their sizes as needed. The specific steps are as follows:

Selecting Modules. As shown in Figure 5a, we offer five basic functional modules, including buttons, knobs, sliders, joysticks, and grips. Users can select the corresponding module and click Show 3D model button, and the module's 3D preview will display in real time on the 3D interface, facilitating visual editing and modification.

Adjusting Parameters. As shown in Figure 5b, users can adjust specific parameters of the modules, such as height, width, and the number of switch circuits, using sliders in the GUI as per their requirements. It is noteworthy that since the shape of the module is fixed, the layout of the circuits is also fixed. Hence, users do not need to design the circuits themselves. By adjusting the proportion of the modules, the circuits will change accordingly. The circuit parameters shown in the Figure 5a can choose the number of switch circuits generated on the module. For example, the press module shown in the Figure 5a can select 0-2 switch circuits.

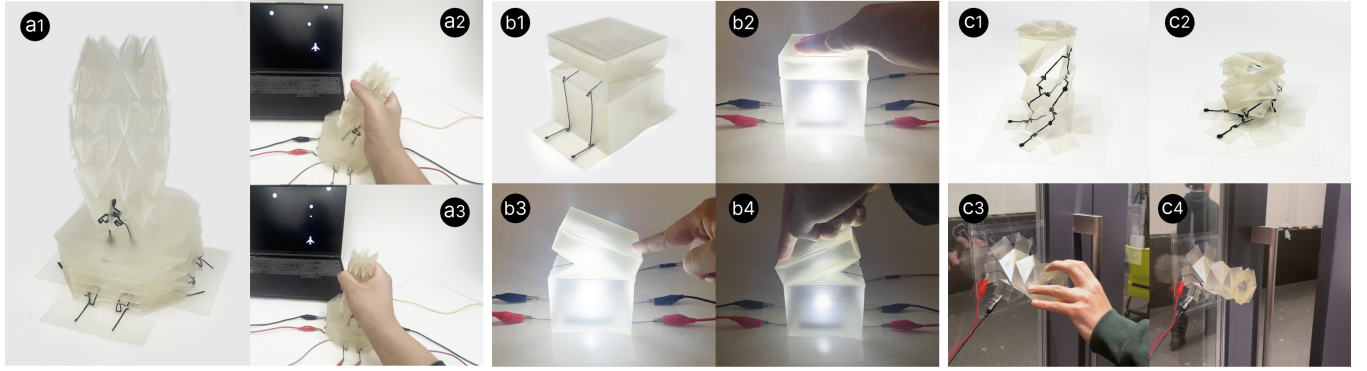


Figure 7: Three applications. (a) a game controller that enables direction control and shooting; (b) a desk lamp that enables the switching of the light and the adjustment of its brightness by means of a circuit; (c) an electronic door lock that can be opened by rotating it to a specific angle

Generating Files. The printing process involves multiple different layers and parameters. Users can click on 'Generate layer files' to create files for processing. As shown in Figure 5c, the plugin automatically generates corresponding layers based on these data. Finally, five different files are produced, including H layer (Figure 5c1), M layer (Figure 5c2), V layer (Figure 5c3), conductive path layer (Figure 5c4, for screen printing), and cutting layer (Figure 5c5, for processing in a laser cutting machine). These layers are automatically generated and exported as processable files based on user-defined parameters.

4.2 Fabrication Process

To provide users with manufacturing guidance, we have summarized the manufacturing process (Figure 6): (a) Import the files generated by the software into a UV printer for printing. (b) Place the material in a laser cutter to process the cutting layer. The laser cutter will trim away the excess parts. (c) Use screen printing to create conductive paths. (d) Submerge the material in a 65 °C water bath to heat and await folding. (e) Dry the prototype, and then connect the modules to the circuit for testing.

5 APPLICATION

5.1 Gamepad

As shown in Figure 7a, we designed a game controller by combining a joystick module and a grip module. We demonstrate how it works through a game. The joystick controls the direction of the spaceship's movement, while the grip controls shooting. There are four switch circuits on the joystick that can detect signal inputs in eight directions, allowing the spaceship to move in eight different directions. The grip module has a switch circuit to detect if it is gripped; when the grip module is clenched, the spaceship shoots. By combining different modules, our method can create a rich tangible input interface.

5.2 Origami Lamp

As shown in Figure 7b, this is a lamp. It is designed using a press module and is equipped with two switch circuits on both the left and right sides. The press module serves as the lampshade, with

a light bulb placed inside. When the user presses down the entire lampshade, and both switch circuits are connected, the light bulb will light up. If the user tilts the lampshade to the right, the brightness of the bulb will continuously increase; tilting it to the left will gradually dim it. Pressing down the entire lampshade again will turn off the bulb. By integrating the circuit within the origami lampshade, a furniture product can be designed where form and function are seamlessly combined.

5.3 Electronic Lock

As shown in Figure 7c, we designed an electronic lock by combining three rotation modules. Each rotation module is equipped with a switch circuit that can detect two states of the knob. By combining the three knobs, we obtained eight different states, allowing users to pre-set a password for unlocking the door. The door will only open when the user turns the electronic lock to a specific state.

6 CONCLUSION

In this paper, we propose a method of rapidly creating tangible interactive interfaces using 4D printing with conductive materials patterning. This method not only enables signal input but also simulates different input methods. We designed five shape modules to mimic various interaction styles, including button, rotation, joystick, slider, and grip modules. Utilizing our approach, designers can quickly create and combine the tangible interfaces they need. Our method also has some limitations, such as a lack of comprehensive assessment of the durability of circuits and structures and the stability of detection methods. Additionally, conductive ink can be printed directly using a UV printer. However, due to the time required to update equipment, we did not directly print conductive paths using a UV printer in this work. We will use a UV printer to directly print circuits in our subsequent research.

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